Task 3 – Adversity Prediction Model

We predict the probability a trade is adverse (LP's PnL < 0) at each horizon $\tau \in \{5,10,15,20,25,30\}$. Data is split chronologically (60% train, 20% validation, 20% test) with an embargo: no training trade's closing mid $M\tau$ occurs after validation start.

Model & Training

We use a weighted ensemble of LightGBM and CatBoost, tuned separately per τ with Optuna (20 trials, early stopping, minimising log-loss). Ensemble weights are inverse validation log-loss. Raw probabilities are Platt-scaled on validation.

Features (52 total, all numeric)

- **Trade & client**: Side, Volume, log_volume, signed_volume, client_code, client_adv_rate (expanding mean of adversity per client, frozen for val/test), client_adv_ratio (rolling 20-trade fraction of informed flow: side=+1 & mid rising OR side=−1 & mid falling).
- **Microstructure**: Spread, spread_to_vol_20, spread_change, spread_skew, tp_m0_norm, side_tp, time_since_last, trade_burst.
- **Price & volatility**: ema_mid_1/5/10, ema_return_1/5/10, vol_5/20/50.
- **Order flow**: signed_volume, cum_signed_volume, ofi_20, volume_ema_5, volume_ratio, client_volume_z.
- **Temporal**: hour, minute, second, hour_sin, hour_cos.
- **Lagged (per client)**: prev_adverse, prev_signed_vol, prev_spread.
- **Interactions**: adv_rate_x_spread, adv_ratio_x_signed_vol, spread_x_vol.

All rolling/lagged features use only past data within each split. Feature order is in feature_order.txt.

Results (averaged over τ)

Split	Accuracy	Precision	Recall	Log-Loss
Train	0.782	0.769	0.745	0.421
Validation	0.741	0.728	0.701	0.489
Test	0.738	0.721	0.695	0.495

Log-loss 0.489 (baseline 0.693) and recall 0.70 indicate good calibration and sufficient sensitivity for externalisation. The model generalises well (test metrics close to validation).